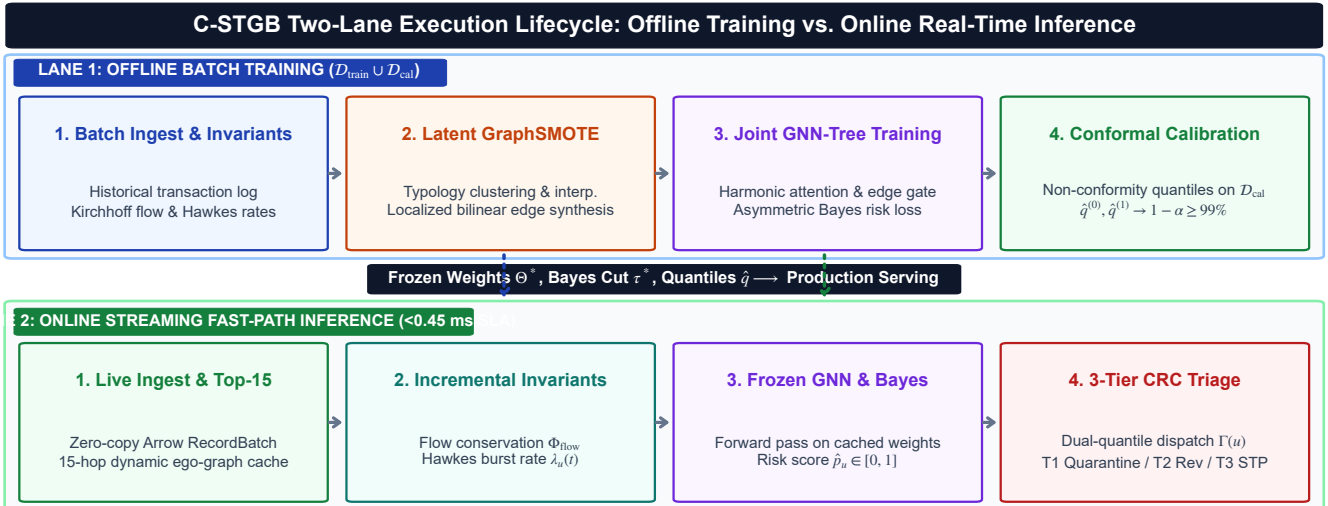

C-STGB Two-Lane Execution Lifecycle: Offline Training vs. Online Real-Time Inference
LANE 1: OFFLINE BATCH TRAINING ($\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{cal}}$)
1. Batch Ingest & Invariants
Historical transaction log
Kirchhoff flow & Hawkes rates
2. Latent GraphSMOTE
Typology clustering & interp.
Localized bilinear edge synthesis
3. Joint GNN-Tree Training
Harmonic attention & edge gate
Asymmetric Bayes risk loss
4. Conformal Calibration
Non-conformity quantiles on $\mathcal{D}_{\text{cal}}$
$\hat{q}^{(0)}, \hat{q}^{(1)} \to 1 - \alpha \geq 99\%$
Frozen Weights $\Theta^*$, Bayes Cut $\tau^*$, Quantiles $\hat{q}$ ⟶ Production Serving
2: ONLINE STREAMING FAST-PATH INFERENCE (<0.45 ms SLA)
1. Live Ingest & Top-15
Zero-copy Arrow RecordBatch
15-hop dynamic ego-graph cache
2. Incremental Invariants
Flow conservation $\Phi_{\text{flow}}$
Hawkes burst rate $\lambda_u(t)$
3. Frozen GNN & Bayes
Forward pass on cached weights
Risk score $\hat{p}_u \in [0, 1]$
4. 3-Tier CRC Triage
Dual-quantile dispatch $\Gamma(u)$
T1 Quarantine / T2 Rev / T3 STP